

“A smart home appliances control system based on digital electronics and GSM network.”

*A project
Submitted in partial fulfillment of the requirements for
the award of the Degree of*

**DIPLOMA OF ENGINEERING
IN
ELECTRICAL AND ELECTRONICS ENGINEERING**

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DECLARATION CERTIFICATE

This is to certify that the work presented in the project entitled “A smart home appliances control system based on digital electronics and GSM network.” in partial fulfillment of the requirement for the award of **Diploma of Engineering in Electrical & Electronics Engineering** of University Polytechnic Birla Institute of Technology, Mesra, Ranchi is an authentic work carried out under my supervision and guidance.

To the best of my knowledge, the content of this project does not form a basis for the award of any previous Degree to any one else.

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Abstract

Abstract—The project proposes an efficient implementation for IoT (Internet of Things) used for monitoring and controlling the home appliances via World Wide Web. Home automation system uses the portable devices as a user interface. They can communicate with home automation network through an Internet gateway, by means of low power communication protocols like Zigbee, Wi-Fi etc. This project aims at controlling home appliances via Smartphone using Wi-Fi as communication protocol and raspberry pi as server system. The user here will move directly with the system through a web-based interface over the web, whereas home appliances like lights, fan and door lock are remotely controlled through easy website. An extra feature that enhances the facet of protection from fireplace accidents is its capability of sleuthing the smoke in order that within the event of any fireplace, associates an alerting message and an image is sent to Smartphone. The server will be interfaced with relay hardware circuits that control the appliances running at home. The communication with server allows the user to select the appropriate device. The communication with server permits the user to pick out the acceptable device. The server communicates with the corresponding relays. If the web affiliation is down or the server isn't up, the embedded system board still will manage and operate the appliances domestically. By this we provide a climbable and price effective Home Automation system.

Today we are living in 21st century where automation is playing important role in human life. Home automation allows us to control household appliances like light, door, fan, AC etc. It also provides home security and emergency system to be activated. Home automation not only refers to reduce human efforts but also energy saving and time efficiency. The main objective of home automation and security is to help handicapped and old aged people who will enable them to control home appliances and alert them in critical situations. This paper put forwards the design of home automation and security system using ARM7 LPC2148 board. The design is based on a standalone embedded system board ARM7 LPC2148 at home. Home appliances are connected to the ARM7 and communication is established between the ARM7 and ARM9 with Bluetooth device. The home appliances are connected to the input / output ports of the embedded system board and their status is passed to the ARM7. We would develop an authentication to the system for authorized person to access home appliances. The device with low cost and scalable to less modification to the core is much

important. It presents the design and implementation of automation system that can monitor and control home appliances via ARM9 S3C2440A board.

In this busy life while away from home we may tend to forget to switch off the AC appliances which cause wastage of energy. Misuse of power energy can be curtailed by automating the devices and appliances. In Home automation household appliances and residential house features like gates, curtains etc are controlled automatically or semi -automatically. Some of the currently available systems provide a view of the house from a web application; but this can cause trouble to the user. Because user must access the web each time he/she wishes to view the status of the home appliances. In this paper a home automation system that uses IR remote, Bluetooth and GSM to control AC appliances using android app is introduced that is easy to use over the traditional method of the switch. Therefore, the motivation behind the development of this system is to let people know about these technologies, and make the system as simple as possible for an ordinary person to understand. The result of this research is the implementation of home automation system which involves control and automation of home appliances through mobile application from remote locations.

The proposed system consists of two main components; the first part is the server (web server), which presents system core that manages, controls, and monitors users' home. Users and system administrator can locally (LAN) or remotely (internet) manage and control system code. Second part is hardware interface module, which provides appropriate interface to sensors and actuator of home automation system. Unlike most of available home automation system in the market the proposed system is scalable that one server can manage many hardware interface modules as long as it exists on WiFi network coverage. System supports a wide range of home automation devices like power management components, and security components. The proposed system is better from the scalability and flexibility point of view than the commercially available home automation systems. Currently in day today's life we can hardly find a house without a home automation system. This project is intended to construct a home automation system that uses any mobile device to control the home appliances. This home automation system is based on IoT. Home automation is very exciting field when it uses new technologies like Internet of Things (IoT). Raspberry pi is credit card size computer. Raspberry pi supports large number of peripherals. Raspberry pi is having different communication media like Ethernet port, HDMI port, USB port, Display Serial Interface, Camera Serial Interface, Bluetooth, Bluetooth low energy. It allows to control number of home appliances simultaneously. Here local server is created on

Raspberry pi. User required to use different mobile devices like smart phones, Laptops, Tablets to operate the home appliances with the help of UI created on web page.

This study presents the design and implementation of a home automation system that focuses on the use of ordinary electrical appliances for remote control using Raspberry Pi and relay circuits and does not use expensive IP-based devices. Common Lights, Heating, Ventilation, and Air Conditioning (HVAC), fans, and other electronic devices are among the appliances that can be used in this system. A smartphone app is designed that helps the user to design the smart home to his actual home via easy and interactive drag & drop option. The system provides control over the appliances via both the local network and remote access. Data logging over the Microsoft Azure cloud database ensures system recovery in case of gateway failure and data record for lateral use. Periodical notifications also help the user to optimize the usage of home appliances. Moreover, the user can set his preferences and the appliances are auto turned off and on to meet user-specific requirements. Raspberry Pi acting as the server maintains the database of each appliance. HTTP web interface and apache server are used for communication between the android app and raspberry pi. With a 5v relay circuit and micro-processor Raspberry Pi, the proposed system is low-cost, energy-efficient, easy to operate, and affordable for low-income houses.

Smart home systems achieved great popularity in the last decades as they increase the comfort and quality of life. Most smart home systems are controlled by smartphones and microcontrollers. A smartphone application is used to control and monitor home functions using wireless communication techniques. We explore the concept of smart home with the integration of IoT services and cloud computing to it, by embedding intelligence into sensors and actuators, networking of smart things using the corresponding technology, facilitating interactions with smart things using cloud computing for easy access in different locations, increasing computation power, storage space and improving data exchange efficiency. In this chapter we present a composition of three components to build a robust approach of an advanced smart home concept and implementation

Literature Review

[1] This paper presents design and implementation of remote-control system by means of GSM cellular communication network. The design integrates the device to be controlled, the microcontroller, and the GSM module so that it can be used for a wide range of applications. Detailed description and implementation of each design element are presented. To verify the principle operation of the M2M design, two home applications are experimentally tested using PC-based environment.[2] The home automation today needs to make use of the latest technological components available. In this paper, we present the design and implementation of a home automation system where communication technologies GSM (Global System for Mobile Communication), Internet, and speech recognition have been used. All these techniques are successfully merged in a single wireless home automation system. This system offers a complete, low cost, powerful and user-friendly way of real-time monitoring and remote control of a house. [3] The long term experience, both positive and negative, of the households we interviewed illustrates four barriers that need to be addressed before home automation becomes amenable to broader adoption. These barriers are high cost of ownership, inflexibility, poor manageability, and difficulty achieving security. Our findings also provide several directions for further research, which include eliminating the need for structural changes for installing home automation, providing users with simple security primitives that they can confidently configure, and enabling composition of home devices.[4] This paper identifies the reasons for this slow adoption and evaluates the potential of Zig-Bee for addressing these problems through the design and implementation of flexible home automation architecture. Device control is a process that is done in the day to day life of mankind. Usually there are a number of devices associated with home and an efficient control of these systems is a tedious task. The rapidly advancing mobile communication technology and the decrease in costs make it possible to incorporate mobile technology into home automation systems. [5] A discussion of the implications of the risk analysis results points to the need for a more general model of security and privacy included in the design phase of smart homes. With such a model of security and privacy in design in place, it will contribute to enforcing system security and enhancing user privacy in smart homes, and thus helping to further realize the potential in such IoT environments. [6] This paper presents a low cost and flexible home control and environmental monitoring system. The proposed system does not require a dedicated server PC with respect to similar systems and offers a novel communication protocol to monitor and control the home environment with more than just

the switching functionality. To demonstrate the feasibility and effectiveness of this system, devices such as light switches, power plug, temperature sensor, gas sensor and motion sensors have been integrated with the proposed home control system. [7] This project aims at controlling home appliances via Smartphone using Wi-Fi as communication protocol and raspberry pi as server system. The user here will move directly with the system through a web-based interface over the web, whereas home appliances like lights, fan and door lock are remotely controlled through easy website. The server will be interfaced with relay hardware circuits that control the appliances running at home. The communication with server allows the user to select the appropriate device. The communication with server permits the user to pick out the acceptable device. The server communicates with the corresponding relays. If the web affiliation is down or the server isn't up, the embedded system board still will manage and operate the appliances domestically. By this we provide a climbable and price effective Home Automation system. [8] This paper discusses about IoT and how it can be used for realizing smart home automation using a micro-controller based Arduino board and Android mobile app. In this paper, two prototypes namely home automation using Bluetooth in an indoor environment and home automation using Ethernet in an outdoor environment are presented. [9] In this paper the working principle of different type of wireless communication techniques such as ZigBee, Wi-Fi, Bluetooth, EnOcean and GSM are studied and their features are compared with each other so the users can choose their own choice of technology to build home automation system. Moreover in this research work the survey of different home automation systems is discussed and their advantages and drawbacks are also highlighted.[10] Ideally, anything that can be connected to a network can be automated and controlled remotely. Smart homes must be artificially intelligent systems that need to adapt themselves based on user actions and surroundings. These systems need to carefully analyse the user needs and the conditions of the surroundings in order to predict future actions and also minimizes user interaction.

Keywords: Home automation, GSM, Smart homes, Internet of Things (IOT), Arduino, NodeMCU, Wireless Fidelity (Wi-Fi)

Introduction

[1] M2M (short for machine-to-machine, man-to-machine or mobile-to-machine) is estimated to get an exponential growth in the coming years. M2M enables the flow of data between machines and machines and ultimately machines and people. Regardless of the type of machine or data, information usually flows from a machine over a network, and then through a gateway to a system where it can be processed and analysed. On other words, M2M allows a machine or device to transmit or receive its data remotely over a network. This makes M2M a very good solution for many problems such as reducing the cost (e.g. the cost of the labours that control or check the machine will be reduced by controlling and checking the machine remotely), and reducing the time of monitoring and controlling the machine (especially when the machine is very far, the time to get to the machine is very long compared to a single click or sending a simple text message). Machines and devices could be vending machines, elevators, pumps, meters, traffic lights, and displays. These need to be constantly monitored, controlled, have information collected from them and sent to them, parameters set and online transactions conducted in order to work effectively. [4] This paper presents a novel, stand alone, low-cost and flexible GSM- ZigBee based home automation system. The architecture is designed to reduce the system's complexity and lower fiscal costs. Hence, the system endeavours not to incorporate complex and expensive components, such as a high end personal computer where possible. The system is flexible and scalable, allowing additional home appliances designed by multiple vendors, to be securely and safely added to the home network with the minimum amount of effort. The system allows home owners to monitor and control connected devices in the home, through a variety of controls, including a ZigBee based remote control. Users may remotely monitor and control their home devices using GSM. [7] Home automation or sensible home may be delineated as introduction of technology within the home atmosphere to provide ease and protection to its occupants. By using the technology of the Internet of Things, the examination and execution of home automation have got additional average. Various wireless technologies which is able to support some sort of remote knowledge transfer, sensing and management like Bluetooth, Wi-Fi and cellular networks are used to enter abundant levels of acumen within the home. Home automation for the older and disabled will offer raised quality of life for persons. It may provide an interface to home appliances or the automation system itself, via telephone line or the internet, to supply management and observance via a smart phone or personal computer. [9]Home automation system is growing rapidly, they are used to provide comfort,

convenience, quality of life and security for residents. Nowadays, most home automation systems are used to provide ease to elderly and disabled people and they reduce the human labor in the production of services and goods. Home automation system can be designed and developed by using a single controller which has the ability to control and monitor different interconnected appliances such as power plugs, lights, temperature and humidity sensors, smoke, gas and fire detectors as well as emergency and security systems. One of the greatest advantage of home automation system is that it can be controlled and managed easily from an array of devices such as smartphone, tablet, desktop and laptop. The rapid growth of wireless technologies influences us to use smartphones to remotely control and monitor the home appliances around the world. Several home automation systems use smartphones to communicate with microcontrollers using various wireless communication techniques such as Bluetooth, GSM, ZigBee, Wi-Fi and EnOcean. Smartphone applications are used to connect to the network so that the authorized users can adjust the setting of system on their personal devices. Different type of home automation systems offer a wide range of functions and services, some of the common features are appliance control, thermostat control, remote control lighting, live video surveillance, monitor security camera, real time text alerts. This paper describes the implementation and working principles of some existing home automation techniques and it compares their cost, speed, real time existence and other functionalities. [10] The top leading industries launch smart hardware and software more often that can be used in home automation systems. However they do not provide an open communication protocol or introduce a way to integrate these different ecosystems. Most of them use proprietary software and set-top boxes or hubs that works well with their own devices, but limits the interoperability with other smart hardware devices that is available in the market. A broad range of appliances in home automation systems use different machine learning and deep learning techniques yet they need manual interaction and do not offer intelligent cross device functionality. Nowadays, the research area in this domain focuses on making smart homes an intelligent decision making systems. This Paper is based on application of Internet of things along with the use of machine learning algorithms. IoT Devices constitute large number of devices and sensors that can monitor/control different physical quantities. With the dawn of Big Data, there is a need for proper automated data storage solutions and cloud based applications to analyse and extract required information from the data. This data can be used for helping or predicting the users action and provide optimized automation.

Classic smart home, internet of things, cloud computing and rule-based event processing, are the building blocks of our proposed advanced smart home integrated compound. Each component contributes its core attributes and technologies to the proposed composition. IoT contributes the internet connection and remote management of mobile appliances, incorporated with a variety of sensors. Sensors may be attached to home related appliances, such as air-conditioning, lights and other environmental devices. And so, it embeds computer intelligence into home devices to provide ways to measure home conditions and monitor home appliances' functionality. Cloud computing provides scalable computing power, storage space and applications, for developing, maintaining, running home services, and accessing home devices anywhere at anytime. The rule-based event processing system provides the control and orchestration of the entire advanced smart home composition.

METHODOLOGY

A. GSM Based Home Automation System Home Automation is done using Global System for Mobile communication [9]. The features of GSM are that it uses the network of the mobile and is battery powered. It consists of two important components which are GSM modem and the microcontroller. The GSM modem is the communication interface between the Home Automation System (HAS) and the end-user. The Microcontroller connects the GSM with both the actuators and sensors of the appliances at home. Here the users are identified by their mobile numbers and these are saved in the system, Electrically Erasable Programmable Read-Only Memory (EEPROM) during configuration. Short Message Service (SMS) sent from authorized user through the mobile phone is received by the GSM modem through the GSM network. The GSM modem is connected to the AT89C55 microcontroller which receives the instructions and decodes it. For serial communication between the GSM modem and the AT89C55 microcontroller, the Relay RS232 is used and these relays in turn are connected to the sensors and the actuators which control or monitor the device.

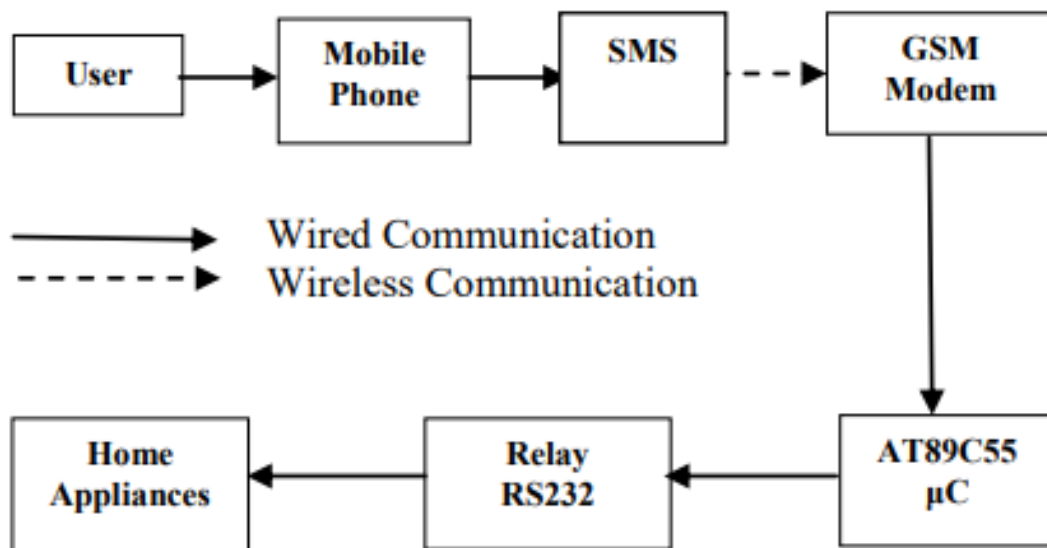


Figure 1. GSM based Home Automation System

C. Bluetooth based Home Automation System Bluetooth is a technology standard which uses short range radio links [11]. Bluetooth based Home Automation system was studied [12]. The system mainly can be divided into two parts: Hardware and Software. Smartphone, Arduino Uno and Bluetooth module HC 06 comprise the Hardware part. Arduino Integrated Development environment (IDE) and Bluetooth terminal comprise the Software part. Bluetooth terminal is used as an app on the smartphone. Using this app, the enduser sends the

command. Once the command is received, it connects onto the Bluetooth module HC 06. The Arduino Uno which is based on ATmega328P Processor then receives the user command in the form of numbers from the interface. These numbers are then designated to the home appliances and the appliances are toggled on or off on receiving the command.

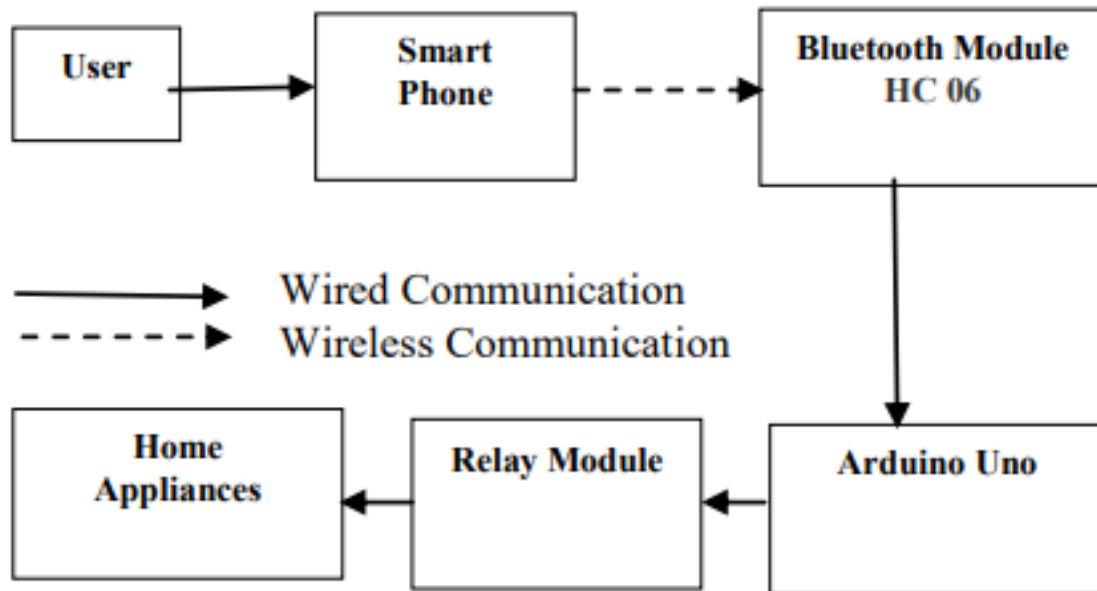


Figure 3. Bluetooth based Home Automation System

D. IoT based Home Automation System Internet of Things comprises of a collection of sensors and actuators which gathers the data from the environment. The data gathered can be further processed to generate information [13]. Mobile Phone GSM Modem AT89C55 μ C Relay RS232 Home Appliances User SMS User Voice Command Kinect Sensor Arduino Uno Relay Module Home Appliances PC Smart Phone Arduino Uno Home Appliances Bluetooth Module HC 06 User Relay Module International Journal of Computer Sciences and Engineering Vol.7(4), Apr 2019, E-ISSN: 2347-2693 © 2019, IJCSE All Rights Reserved 956 This system allows the users to remotely control the appliances in their absentia at the home. IoT based Home Automation system was implemented [14]. The Main components of the system are Arduino Mega, Wi-Fi Module ESP8266, Relay Board. Many sensors and equipments are used to monitor or supervise the home appliances. Virtuino, an Android application is used to control the smart home. Arduino Mega microcontroller interacts with the Virtuino application. The Wi-Fi module, sensors are connected to the microcontroller. Once the relay board receives the input from the microcontroller, the action can be seen on the home appliances. It provides a more reliable security. However, security becomes a major

concern for IoT due to the enormous amount of data that is passing through these systems.

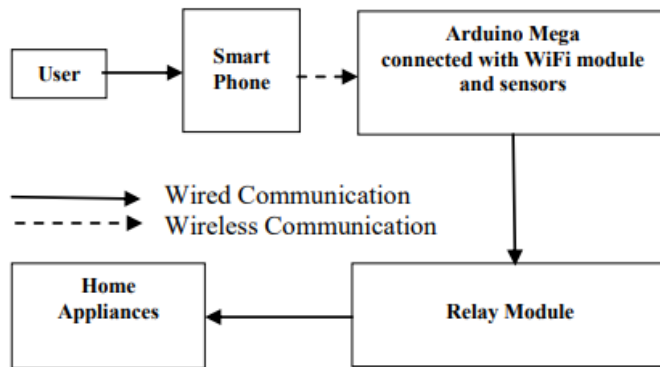


Figure 4. IoT based Home Automation System

Comparision

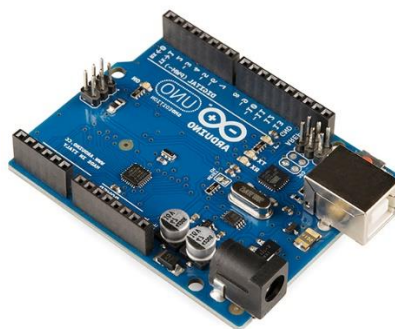
The following table gives the comparison between the different Home Automation Systems with respect to its Speed, Cost and Availability. In GSM based Home Automation system, communication between the user and the appliances is done through SMS. The advantages of GSM based Home Automation is it is easy to implement and is relatively at a lower cost. It is easily understandable as it is based on a simple concept of SMS. It is a global communication technology; therefore, it is accessible anywhere. The limitations are that it exhibits low performance in case of network congestion and limitations in coverage in terms of rural areas and the cost depends on the distance between the user and the home appliance. In Voice Recognition based Home Automation system, communication between the user and the appliances can take place through voice sensors or Android applications. The advantages of Voice Recognition based Home Automation are that it is extremely helpful for people with physical constraints. The limitations are it is always better to use a slider for adjusting the volume rather than specifying the volume count. In case of Voice Recognition Home Automation system using an Android App, it is not completely free handed as we still need to press a button and then only give the command, if it becomes completely handsfree, then specific hardware has to be installed which might make the whole system expensive. In case of a noisy environment, it might not show adequate performance. In Bluetooth based Home Automation System, communication between the user and the appliances occurs through the Bluetooth terminal. The advantages of Bluetooth based Home Automation are that it is suitable for a shorter range, user adaptable, ease of deployment, free ad-hoc networking and data transmission control. The limitations are it is comparatively slow, the data rate is low and it has a constraint of a short distance of up to 10 meters. Once the user is out of the range, no

communication can take place between the user and the system. In IoT based Home Automation system, communication between the user and the appliances take place through the Internet. The advantages are the end-users can remotely access the smart home. The limitations are the end-devices can reveal the user's identity, there is always carriage of data between the smart home and the user as long as the application is concerned; This can lead to intrusion as in many other cases such as social media etc. [22]. With 5G implementation, there is improved security and reliability. However, since 5G uses millimetre waves, it is prone to be affected by atmospheric parameters also there can be an occurrence of penetration loss leading to system inefficiency.

Syatem	Cost	Speed	Availability
GSM using microcontroller	High	Slow	High
Bluetooth using Arduino	Low	Fast	Low
IoT	Low	Fast	High

Components and Sensors:

1. Arduino Uno:



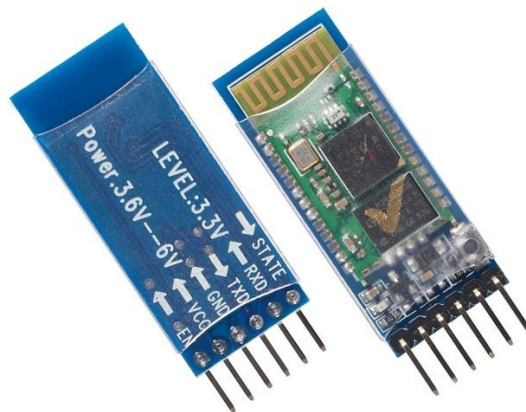
The Arduino Uno stands as a flagship microcontroller board renowned for its versatility and user-friendly design. Developed by the Arduino company, it has become a cornerstone in the

world of electronics and DIY projects. The Uno features an ATmega328P microcontroller, clocked at 16 MHz, offering a robust platform for embedded systems and rapid prototyping.

Equipped with a diverse array of digital and analog input/output pins, the Arduino Uno accommodates a broad range of sensors, actuators, and communication modules. Its open-source nature fosters a thriving community, sharing a vast repository of code and project ideas. The Uno's simplicity in programming, using the Arduino Integrated Development Environment (IDE), makes it accessible to both beginners and seasoned developers.

Notably, the board's popularity stems from its seamless integration with various shields—additional boards that expand its functionalities. Whether employed in robotics, home automation, or educational settings, the Arduino Uno remains a staple, empowering enthusiasts and professionals alike to bring their electronic creations to life. Its enduring influence underscores the democratization of electronics, enabling individuals worldwide to explore and innovate within the realm of embedded systems and IoT applications.

2. Bluetooth Module:



The Bluetooth module establishes a short-range wireless communication link between the Arduino and user devices. This connectivity allows for manual control and configuration through a dedicated mobile app. Users can remotely interact with the system, adjusting settings, and monitoring the status of connected devices.

3. WiFi Module:

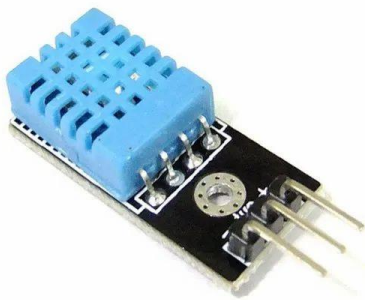


A WiFi module is a compact electronic component that integrates with devices to provide wireless internet connectivity. It serves as a bridge between a device, such as a microcontroller or sensor, and the vast realm of the internet. WiFi modules operate on the IEEE 802.11 standard, enabling seamless communication within local area networks (LANs) and connecting devices to the internet.

Typically, these modules come equipped with embedded processors and communication protocols, allowing them to handle the complexities of wireless communication. They support various security standards, such as WEP, WPA, and WPA2, ensuring secure data transmission over the airwaves. WiFi modules are widely utilized in IoT devices, home automation systems, and other applications where wireless connectivity is paramount.

One notable example is the ESP8266, a popular WiFi module known for its affordability and versatility. It empowers hobbyists and developers to imbue their projects with internet capabilities, facilitating remote control, data transmission, and real-time monitoring. As our world becomes increasingly interconnected, WiFi modules play a pivotal role in shaping the landscape of smart devices, enabling them to communicate, share data, and operate seamlessly in the digital age.

4. Temperature Sensor:



A temperature sensor is a crucial component in electronics and automation, designed to measure the ambient temperature of its surroundings. These sensors operate on various principles, with the most common types being thermocouples, thermistors, and infrared sensors.

Thermocouples generate a voltage proportional to the temperature difference between two junctions, while thermistors use the temperature-dependent resistance of semiconductor

materials. Infrared sensors, on the other hand, detect the infrared radiation emitted by an object to calculate its temperature.

Temperature sensors find extensive application in diverse fields, including weather monitoring, industrial processes, and consumer electronics. In the context of your home automation project, a temperature sensor can play a pivotal role in climate control. For example, it can trigger heating or cooling systems based on predefined temperature thresholds, ensuring a comfortable living environment.

One commonly used temperature sensor is the DS18B20, known for its digital output, high precision, and compatibility with various microcontrollers. Whether embedded in weather stations or integrated into smart thermostats, temperature sensors are essential tools for gathering environmental data and enabling precise control in a wide array of technological applications.

5. Light Sensor:



A light sensor, also known as a photodetector or photoresistor, is a fundamental component in electronic systems designed to detect and measure the intensity of light in its surroundings. Light sensors operate on the principle of varying electrical resistance in response to incident light, making them essential for applications ranging from automatic lighting control to photography.

One common type of light sensor is the photoresistor, which exhibits a change in resistance based on the amount of incident light. As light intensity increases or decreases, the sensor's resistance alters proportionally, allowing it to convert light into an electrical signal. Another variant is the photodiode, which generates a current when exposed to light, converting photons into an electrical charge.

Light sensors have diverse applications, contributing to energy-efficient lighting systems, photography exposure control, and even ambient light adjustment in electronic displays. In the context of your home automation project, a light sensor can be utilized to automatically adjust the brightness of indoor lighting based on natural light levels, contributing to energy savings and creating a more comfortable living environment. With their versatility and wide-ranging applications, light sensors are integral components in modern electronics and automation, enhancing both efficiency and user experience.

6. PIR (Passive Infrared) Sensor:



A Passive Infrared (PIR) sensor is a key component in motion detection technology, widely used in various applications, including security systems and home automation. The PIR sensor detects infrared radiation emitted by objects in its field of view, sensing changes in temperature and movement. Unlike active sensors, PIR sensors do not emit energy; instead, they passively respond to the infrared radiation naturally emitted by living beings and other objects.

The core principle of a PIR sensor lies in its pyroelectric material, which generates an electric charge in response to temperature variations. When an object, especially one with a different temperature than its surroundings, moves across the sensor's field, it triggers a change in the generated charge, indicating motion.

In home automation, PIR sensors play a crucial role in enhancing security and energy efficiency. For example, when integrated into a lighting system, a PIR sensor can detect human presence in a room and automatically turn on the lights, providing illumination only when needed. Similarly, in security applications, PIR sensors can trigger alarms or surveillance cameras in response to detected motion, contributing to effective intrusion detection.

The simplicity, reliability, and effectiveness of PIR sensors make them invaluable in creating responsive and energy-efficient systems, offering a seamless integration of technology for both convenience and safety in various environments.

7. Ultrasonic Motion Sensor:



An ultrasonic motion sensor is a technological marvel that utilizes ultrasonic waves to detect movement and measure distance. It typically consists of a transmitter that emits ultrasonic pulses and a receiver that detects the reflected waves. By calculating the time taken for the waves to travel to an object and back, the sensor determines the distance and identifies motion in its field of view.

The advantages of ultrasonic motion sensors lie in their accuracy and versatility. They operate effectively in various lighting conditions and are less prone to false triggers caused by environmental factors like temperature changes or ambient light. These sensors are commonly employed in occupancy sensing, security systems, and robotics. For instance, in your home automation project, an ultrasonic motion sensor could be used to activate lights or adjust the thermostat when someone enters a room, contributing to energy efficiency and user convenience.

Overall, ultrasonic motion sensors represent a sophisticated and reliable technology, offering precise motion detection and distance measurement for a diverse range of applications.

8. Relay



A relay is an electromechanical device that plays a crucial role in controlling electrical circuits by using a low-power signal to manage a higher-power circuit. Its fundamental purpose is to act as a switch, enabling the control of one electrical circuit by opening or closing contacts in another circuit. The relay consists of a coil, an armature, a set of contacts, and a spring. When an electric current flows through the coil, it generates a magnetic field that attracts the armature, causing the contacts to either open or close, depending on the relay type. Relays come in various configurations, including electromagnetic relays, solid-state relays, and thermal relays, each serving specific purposes in different applications.

Electromagnetic relays, the most common type, utilize an electromagnet to mechanically operate the switch. As the coil receives an electric current, it produces a magnetic field that pulls the armature, either breaking or establishing the connection in the contacts. These relays are widely used in applications like automation systems, industrial controls, and automotive circuits. On the other hand, solid-state relays (SSRs) employ semiconductor devices, such as thyristors or transistors, to perform the switching operation without any moving parts. SSRs are preferred in situations where high-speed and noiseless operation is crucial, like in microprocessor-controlled systems or when interfacing with sensitive electronics.

Thermal relays, also known as overload relays, protect electrical circuits from overheating by monitoring the current flow. In case of an excessive current, the relay triggers a response to interrupt the circuit, preventing damage to the equipment. This is particularly essential in applications like electric motors, where protection against overload is vital.

Relays find widespread use in numerous industries due to their versatility and reliability. In the field of power systems, they safeguard transformers, generators, and other critical components from faults by quickly isolating faulty sections. In the automotive sector, relays control various electrical components such as lights, fans, and motors. Moreover, in home automation, relays enable the remote control of lights, appliances, and heating systems. Their ability to isolate low-voltage control circuits from high-voltage power circuits ensures the safety of users and the longevity of electronic devices.

Working Principle:

1. Data Acquisition:

The home automation system starts with the continuous collection of data from various sensors. The temperature sensor monitors the ambient temperature, the light sensor gauges light levels, and the motion detectors, both PIR and ultrasonic, identify any movement in their respective areas. These sensors feed real-time data to the Arduino, creating a comprehensive snapshot of the home environment.

2. Data Processing:

The Arduino processes the incoming sensor data using pre-programmed logic. This logic, defined in the system's code, dictates how the system responds to different sensor inputs. For example, if the temperature exceeds a predefined threshold, the Arduino may trigger the air conditioning system to cool the space. Similarly, if motion is detected by the PIR sensor, the system might turn on lights and send a notification to the user's device.

3. User Interaction:

The Bluetooth module enables user interaction with the system through a mobile app. Users can manually control connected devices, set preferences, and receive real-time updates about the status of the home. The app acts as a user-friendly interface, allowing for seamless communication with the Arduino and providing an avenue for personalized control.

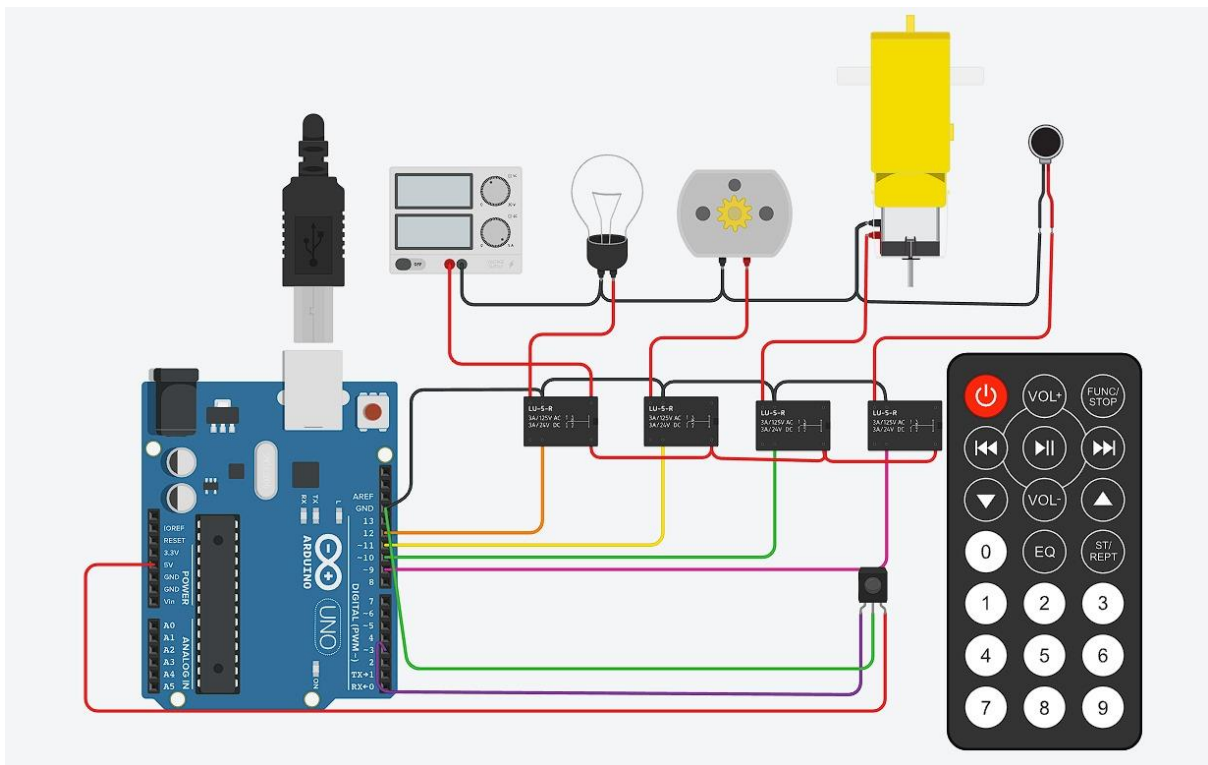
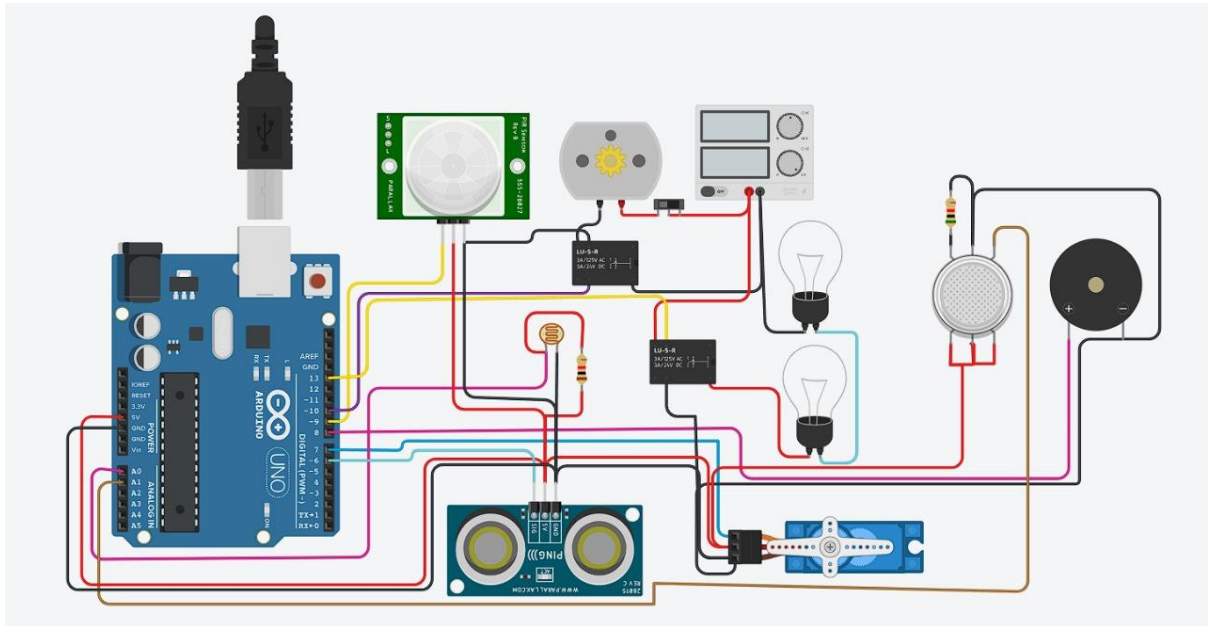
4. Internet Connectivity:

The WiFi module extends the project's reach by connecting the Arduino to the internet. This connectivity is pivotal for remote monitoring and control. Users can access the system from anywhere globally, allowing for real-time adjustments, status checks, and notifications.

5. Automation and Control:

Based on the processed sensor data and user-defined rules, the Arduino automates various tasks. For instance, it can schedule lighting based on the time of day, adjusting brightness levels to simulate natural lighting conditions. Additionally, the system can optimize energy usage by adjusting the thermostat based on occupancy and ambient temperature, contributing to a more sustainable and cost-effective living space.

WORKING MODEL



Controlling home automation using Arduino in Tinkercad is a fascinating project that combines hardware and software to create a simulated smart home environment. The basic setup involves integrating various components virtually, such as infrared (IR) modules for

remote control, relays to operate devices, and the Arduino for processing signals. By writing code, you can interpret remote signals and trigger corresponding actions on the connected devices.

Hardware Components:

1. **Arduino Board:** The core of the setup, an Arduino board processes commands and controls connected devices.
2. **Infrared (IR) Modules:** These modules receive signals from a remote control. In the Tinkercad simulation, you can use a virtual IR module to emulate the reception of IR signals.
3. **Relays:** Relays act as switches, allowing the Arduino to control high-voltage devices like lights or appliances. Tinkercad offers simulated relays for this purpose.
4. **Devices:** Simulated devices like lights or fans represent what you'd control in a real home automation scenario.

Software Components:

1. **Arduino Code:** Write code to program the Arduino. This involves libraries to interface with IR modules, defining actions based on received signals, and controlling relays.
2. **IR Remote Library:** In Tinkercad, you can use the virtual IR Remote library to simulate the signals sent by a remote control. This allows you to test the interaction between the IR module and the Arduino.

Setting Up the Simulation:

1. **Connect Components:** In Tinkercad, drag and drop components onto the virtual workspace. Connect the IR module to the Arduino, and connect relays to the devices you want to control.
2. **Write Arduino Code:** Develop code to read signals from the IR module and define actions based on those signals. For example, pressing "Power" on the virtual remote could turn on a simulated light.

CODE

```
#include <IRremote.hpp>

const int rcvPin=3;

int led=12;

int mot=11;

int moto=10;

int vib=9;


IRrecv irrecv(rcvPin);


void setup()
{
  Serial.begin(9600);

  //IrReceiver.begin(rcvPin, true);

  irrecv.enableIRIn(); // Start the receiver

  pinMode(led, OUTPUT);

  pinMode(mot, OUTPUT);

  pinMode(moto, OUTPUT);

  pinMode(vib, OUTPUT);
}


void loop()
{
```

```

if(IrReceiver.decode()){

    auto value= IrReceiver.decodedIRData.decodedRawData;


    switch(value)  {

        case 4010852096: //keypad no '1'

            digitalWrite(led,HIGH);

            Serial.println(1);


    }


    switch(value)

    {

        case 3994140416: //keypad no '2'

            digitalWrite(mot,HIGH);

            Serial.println(2);

    }

    switch(value)

    {

        case 3977428736: //keypad no '3'

            digitalWrite(moto,HIGH);

            Serial.println(3);

    }

    switch(value)

    {

```

```

        case 3944005376: //keypad no '4'

digitalWrite(vib,HIGH);

        Serial.println(4);
    }
switch(value)
{
        case 3927293696: //keypad no '5'

digitalWrite(led,LOW);

        Serial.println(5);
    }switch(value)
    {
        case 3910582016: //keypad no '6'

digitalWrite(mot,LOW);

        Serial.println(6);
    }switch(value)
    {
        case 3877158656: //keypad no '7'

digitalWrite(moto,LOW);

        Serial.println(7);
    }switch(value)
    {
        case 3860446976: //keypad no '8'

digitalWrite(vib,LOW);

        Serial.println(8);
    }

```

```

    }

    switch(value)

    {

        case 4278238976: //keypad no 'Power'

            digitalWrite(led,LOW);

            digitalWrite(mot,LOW);

            digitalWrite(moto,LOW);

            digitalWrite(vib,LOW);

            Serial.println("power");

        }

        IrReceiver.resume();

    }

}

```

Benefits:

1. Convenience:

Remote Access: Homeowners can control and monitor various devices and systems remotely, providing unparalleled convenience. This includes adjusting thermostat settings, turning lights on or off, and even checking security cameras from a mobile device.

2. Energy Efficiency:

Smart Thermostats: Home automation systems often include smart thermostats that learn user preferences and adjust heating and cooling accordingly, optimizing energy usage.

Lighting Control: Automated lighting systems can be programmed to turn off lights when a room is unoccupied or adjust brightness levels based on natural light conditions, contributing to energy savings.

3. Security:

Surveillance Systems: Integrated security cameras and motion sensors enhance home security. Users can receive real-time alerts and monitor their property remotely.

Smart Locks: Home automation allows for keyless entry and remote control of door locks, adding an extra layer of security.

4. Safety:

Automated Alerts: Smart sensors can detect issues like water leaks, smoke, or carbon monoxide, triggering automated alerts to homeowners, potentially preventing significant damage or harm.

5. Customization and Personalization: Adaptive Environments:

Home automation allows users to customize their living spaces. Lighting, temperature, and entertainment systems can be tailored to individual preferences, creating a personalized and comfortable environment.

6. Time Savings:

Automated Tasks: Repetitive tasks, such as adjusting blinds or turning off appliances, can be automated, saving time and effort for homeowners.

7. Increased Home Value:

Appeal to Buyers: A well-integrated home automation system can increase the resale value of a property. Many homebuyers are attracted to the added convenience, security, and energy efficiency offered by smart homes.

8. Peace of Mind:

Remote Monitoring: Homeowners can remotely monitor their homes, ensuring that everything is secure and functioning correctly, providing peace of mind, especially when away.

9. Accessibility for All:

Assistance for Elderly or Disabled: Home automation can make living spaces more accessible for elderly or disabled individuals. Smart devices can be adapted to assist with daily tasks and enhance independence.

10. Cost Savings:

Energy savings and preventive maintenance are two pivotal aspects of home automation that significantly enhance the overall efficiency and longevity of a household. Automated systems designed to optimize energy usage represent a cornerstone in the evolution of smart homes. These systems often incorporate advanced technologies, such as smart thermostats, energy-efficient lighting controls, and smart appliances. Through real-time monitoring and intelligent algorithms, they adapt to user preferences and environmental conditions, ensuring that energy is utilized judiciously. For instance, smart thermostats learn the daily routines of occupants and adjust heating or cooling accordingly, contributing not only to a more comfortable living environment but also to substantial energy savings. By automating the control of lights, appliances, and HVAC systems, homeowners can effortlessly reduce energy consumption when certain areas are unoccupied or during periods of lower demand, reflecting positively on utility bills and the overall environmental footprint.

HISTORY

The concept of home automation has roots in early 20th-century science fiction, where authors envisioned homes with advanced technological systems controlled at the push of a button. However, the practical implementation of such ideas began to take shape in the mid-20th century.

1930s - The Emergence of Labor-Saving Devices:

The 1930s saw the introduction of labor-saving devices like dishwashers and refrigerators. These innovations marked the beginning of the mechanization of household tasks, setting the stage for more sophisticated automation in the future.

1960s - The Birth of Home Automation:

The term "home automation" was coined in the early 1960s. The ECHO IV, developed by engineer Jim Sutherland in 1966, is often considered the first home automation system. It could control appliances and was showcased in the popular science magazine "Popular Mechanics."

The Rise of X10 (1970s - 1980s):

1

1970s - X10 Protocol:

The X10 protocol, introduced in the 1970s, became a milestone in home automation history. It allowed compatible devices to communicate over existing power lines, enabling the remote control of lights and appliances. X10 paved the way for the development of various home automation systems.

1980s - Commercialization of X10:

In the 1980s, X10 technology became commercially available, leading to a surge in the popularity of home automation. Consumers could purchase X10 modules to control lighting,

heating, and other devices remotely. However, X10 faced challenges such as signal reliability issues and limited functionality.

A Shift Towards Standardization (1990s):

1990s - Emergence of Industry Standards:

As home automation gained traction, the need for industry standards became apparent. The Home Automation and Networking Association (HANA) was formed in 1992 to address this. However, despite efforts, the lack of a unified standard hindered widespread adoption.

Late 1990s - Rise of Home Automation Companies:

The late 1990s saw the emergence of companies dedicated to home automation. Crestron, founded in 1972 but gaining prominence in the late '90s, focused on high-end automation solutions. Meanwhile, companies like Leviton and Smarthome started offering a broader range of affordable home automation products.

Internet Era and Smart Homes (2000s - Early 2010s):

2000s - Internet Connectivity:

The 2000s marked a significant shift with the integration of the internet into home automation systems. This allowed users to control devices remotely via web interfaces. Companies like Control4 and AMX started providing more sophisticated solutions for audio-visual control, lighting, and security.

Late 2000s - Smart Home Hubs: Smart home hubs, such as the SmartThings Hub, started to gain popularity. These hubs served as central controllers, connecting various smart devices and allowing users to create more complex automation scenarios.

2010s - Proliferation of IoT Devices:

The 2010s witnessed the proliferation of Internet of Things (IoT) devices. Companies like Nest introduced smart thermostats, while Philips Hue brought smart lighting to mainstream consumers. The rise of voice-controlled assistants like Amazon Alexa and Google Assistant further simplified home automation.

Present Day (Mid-2010s - Present):

Mid-2010s - Integration of AI and Machine Learning:

The integration of artificial intelligence (AI) and machine learning (ML) has brought about more intelligent and adaptive home automation. Systems can now learn user preferences, anticipate needs, and optimize energy consumption.

Expansion of Ecosystems:

Major tech companies, including Apple, Google, and Amazon, have entered the home automation space, leading to the development of comprehensive ecosystems. Smart home devices are designed to seamlessly integrate with these ecosystems, providing users with a unified and user-friendly experience.

Edge Computing and Local Processing:

A notable trend in recent years is the emphasis on edge computing and local processing. Smart devices are becoming more capable of processing data locally, reducing reliance on cloud services and enhancing privacy and response times.

Sustainability and Energy Efficiency:

Modern home automation solutions focus on sustainability and energy efficiency. Smart thermostats, solar panels, and energy monitoring systems help users minimize their environmental impact and reduce energy bills.

Continued Innovation:

As of the present day, home automation continues to evolve rapidly. Innovations include advanced security systems, robotic vacuum cleaners, and even more intuitive voice-controlled interfaces. The market is dynamic, with ongoing research and development to bring new, cutting-edge technologies to smart homes.

Future Perspectives:

The future of home automation holds tremendous promise, with advancements in technology poised to transform residences into even smarter, more connected, and efficient living spaces. Several key trends and innovations are shaping the future landscape of home automation.

1. 5G Connectivity:

The rollout of 5G networks will significantly impact home automation by providing faster and more reliable connectivity. With lower latency and higher data transfer rates, 5G will enable seamless communication between devices, supporting real-time interactions and enhancing the overall responsiveness of smart home systems. This is particularly crucial for applications like video streaming, virtual reality, and augmented reality, which could become integral parts of smart home experiences.

2. Artificial Intelligence (AI) Integration:

The integration of AI into home automation systems is set to revolutionize the way smart homes operate. AI algorithms will learn user preferences over time, adapting to individual behaviors and optimizing automation routines. Smart homes will become more anticipatory, understanding and predicting user needs, whether it's adjusting the thermostat, optimizing energy usage, or suggesting automation scenarios based on daily patterns.

3. Edge Computing and Local Processing:

There is a growing emphasis on edge computing and local processing in home automation. Devices are becoming more capable of processing data locally, reducing dependence on cloud services. This not only enhances privacy by minimizing data transfer to external servers but also improves response times for critical automation tasks. Edge computing enables smart devices to function autonomously, even in the absence of an internet connection.

4. Interoperability and Standardization:

As the home automation market continues to expand, the need for interoperability and standardization becomes paramount. Efforts by industry alliances and standards organizations aim to create unified frameworks that allow devices from different manufacturers to seamlessly work together. This will simplify the user experience, reduce fragmentation, and encourage the development of a more comprehensive and cohesive smart home ecosystem.

5. Voice and Gesture Control Evolution:

Voice and gesture control are likely to evolve significantly in the future. Natural language processing advancements will enable more intuitive and context-aware voice commands, making interactions with smart home systems even more conversational. Additionally, gesture recognition technologies could offer a hands-free alternative for controlling devices, providing an added layer of convenience.

6. Energy Harvesting and Sustainability:

The future of home automation will witness increased emphasis on energy harvesting and sustainability. Smart devices may incorporate energy harvesting technologies, such as solar or kinetic energy harvesting, to reduce reliance on traditional power sources. This aligns with the growing global focus on sustainable living practices, making smart homes more eco-friendly and energy-efficient.

7. Augmented Reality (AR) and Virtual Reality (VR) Integration:

The integration of AR and VR technologies into home automation could transform the way users interact with their smart environments. AR overlays could provide real-time information about connected devices, and VR could offer immersive experiences for remote monitoring or virtual walkthroughs of smart home setups. These technologies have the potential to enhance user engagement and troubleshooting processes.

8. Health and Wellness Monitoring:

Home automation is likely to play an increasing role in health and wellness monitoring. Smart homes may integrate advanced sensors and wearables to monitor vital signs, sleep patterns, and overall well-being. The data collected could be analyzed to provide insights into health trends, potentially alerting users and healthcare professionals to early signs of health issues.

9. Security Innovations:

Continued advancements in security technologies will fortify smart homes against cyber threats and physical intrusions. Enhanced encryption, biometric authentication, and AI-powered threat detection systems will be integral components of future smart home security solutions, providing homeowners with peace of mind and robust protection.

10. Customization and Personalization:

The future of home automation will place a premium on customization and personalization. Smart systems will become increasingly intuitive, adapting to individual preferences without the need for explicit user input. This level of personalization ensures that automation enhances the user experience rather than being a one-size-fits-all solution.

In conclusion, the future trajectory of home automation holds the promise of an extraordinary era marked by continual innovation and seamless integration. As technological advancements mature and become increasingly accessible, smart homes are poised to evolve into highly sophisticated ecosystems that revolutionize the way individuals interact with their living spaces. This evolution is expected to significantly enhance various facets of daily life, including convenience, energy efficiency, security, and overall quality of life on a global scale.

The collaborative efforts of leading tech companies, ongoing standardization initiatives, and a burgeoning demand for intelligent living environments are key driving forces propelling the evolution of home automation. Technological convergence, where various smart devices seamlessly communicate and collaborate, will likely be a hallmark of future smart homes.

Software Used

Tinkercad: A Comprehensive Exploration

Tinkercad stands as a versatile and accessible online platform for creating, simulating, and sharing virtual models and circuits. Developed by Autodesk, it caters to a diverse audience, from beginners exploring the basics of 3D design and electronics to advanced users prototyping complex projects. This detailed exploration delves into the various aspects of Tinkercad, covering its features, applications, ease of use, educational benefits, and the integration of 3D design and electronics.

I. Core Features and Functionality

1. 3D Design:

At its core, Tinkercad is renowned for its 3D design capabilities. Users can create detailed and intricate 3D models using a simple drag-and-drop interface. The library of shapes, letters, numbers, and pre-designed components allows for the seamless construction of diverse objects.

2. Circuit Simulation:

Beyond 3D design, Tinkercad's circuit simulation feature sets it apart. With an extensive component library covering various electronic elements, users can design, simulate, and test circuits in a virtual environment. This functionality is particularly valuable for electronics enthusiasts, hobbyists, and students learning about circuitry.

3. Code Blocks and Arduino Integration:

Tinkercad incorporates a Code Blocks feature that enables users to program Arduino-based projects using a visual coding interface. This simplifies the coding process, making it accessible to those with little to no coding experience. The platform seamlessly integrates with Arduino, allowing users to combine hardware and software in a single simulation.

4. Simulation Environment:

The simulation environment in Tinkercad provides real-time feedback on designs and circuits. Users can observe how their projects behave, identify potential issues, and test different scenarios without the need for physical prototypes. This not only saves time but also enhances the prototyping and testing phases of a project.

5. Collaboration and Sharing:

Tinkercad facilitates collaboration through its cloud-based nature. Multiple users can work on a project simultaneously, making it an ideal tool for group projects or educational settings. Users can share their designs and circuits publicly or privately, fostering a collaborative and supportive community.

II. Ease of Use

1. User-Friendly Interface:

Tinkercad's user interface is designed with simplicity in mind. The drag-and-drop functionality, along with an intuitive toolbar, makes it accessible even for beginners. Users can navigate the platform effortlessly, creating 3D designs and electronic circuits with minimal learning curve.

2. Learning Resources:

Tinkercad provides a wealth of learning resources, including tutorials, videos, and community forums. These resources guide users through the platform's features, helping them grasp 3D design principles, understand electronics, and learn programming concepts. The platform's emphasis on education contributes to its popularity in classrooms and online learning environments.

3. Code Blocks for Programming:

The Code Blocks feature simplifies programming by allowing users to drag and snap together visual code blocks. This visual approach makes coding more accessible to beginners, fostering an understanding of programming logic without the need to memorize syntax. Advanced users can seamlessly transition to traditional coding environments with the skills acquired on Tinkercad.

4. Instant Feedback and Iteration:

One of Tinkercad's strengths lies in providing instant feedback. As users design and simulate, they can quickly identify any issues or areas for improvement. This iterative process encourages experimentation and refinement, enhancing the learning experience for users of all skill levels.

III. Applications

1. Education:

Tinkercad has become a staple in educational settings, from elementary schools to universities. Its user-friendly interface and comprehensive learning resources make it an excellent tool for introducing students to 3D design, electronics, and programming. Teachers can create assignments, monitor student progress, and provide valuable feedback in a virtual environment.

2. Prototyping:

For makers, hobbyists, and professionals, Tinkercad serves as an invaluable prototyping tool. It allows users to visualize and test concepts before investing time and resources in physical prototypes. The combination of 3D design and circuit simulation simplifies the prototyping process, enabling users to iterate rapidly and refine their projects.

3. DIY Projects:

Tinkercad caters to the thriving do-it-yourself (DIY) community. Users can design custom 3D-printable objects, simulate electronic projects, and share their creations with a global community. The platform's versatility accommodates a wide range of DIY projects, from simple designs to intricate inventions.

4. Professional Design:

Professionals in various industries leverage Tinkercad for initial design exploration and visualization. Architects, engineers, and product designers can quickly sketch out ideas, assess feasibility, and collaborate with team members in a collaborative online environment. Tinkercad's utility extends beyond educational and hobbyist purposes into the realm of professional design and prototyping.

IV. Integration of 3D Design and Electronics

1. Seamless Integration:

Tinkercad seamlessly integrates 3D design and electronics within a single platform. Users can create a 3D model and embed electronic components directly into it, resulting in projects that combine physical form and functional circuitry. This integration provides a holistic approach to product design, allowing users to consider both the aesthetics and functionality of their creations.

2. Component Library:

Tinkercad's extensive component library includes a wide array of electronic elements, ranging from basic resistors and LEDs to complex integrated circuits. Users can select, place, and connect these components within their 3D designs, fostering a unified approach to designing interactive and visually appealing projects.

3. Arduino Integration:

The incorporation of Arduino compatibility enhances Tinkercad's capabilities. Users can design Arduino-based projects, program them using Code Blocks or traditional code, and simulate their behavior. This integration streamlines the process of combining hardware and software, providing a comprehensive environment for Arduino enthusiasts.

4. Realistic Simulations:

The realistic simulations in Tinkercad allow users to visualize how electronic components interact within the context of a 3D design. This feature is particularly beneficial when designing projects that involve both physical structure and embedded electronics. Users can assess the spatial relationships and functional aspects of their designs simultaneously.

V. Future Developments and Challenges

1. Future Developments:

Tinkercad continues to evolve with updates and new features. As technology advances, there is potential for enhanced realism in simulations, expanded libraries of components, and improved collaboration tools. The platform may further integrate emerging technologies, such as augmented reality, to provide users with even more immersive design and simulation experiences.

2. Challenges:

While Tinkercad excels in accessibility and ease of use, some users may find its capabilities limiting for highly complex or specialized projects. Advanced users seeking more sophisticated features may eventually migrate to dedicated 3D design or CAD software..

As Result

This paper has reviewed the existing state of home automation systems, and identified and discussed five areas that have hindered consumer adoption of such technologies. Briefly, the areas include: the complexity and expense of the architectures adopted by existing systems, the intrusiveness of the system installations, the lack of interoperability between different home automation technologies, and the lack of interoperability between systems developed by different manufacturers that utilize the same technology. Interface inflexibility and the inconsistent approaches adopted towards security and safety are also problems. A novel architecture for a home automation system is proposed and implemented, using the relatively new communication technology ZigBee. The use of ZigBee communications technology helps lower the expense of the system and the intrusiveness of the respective system installation. The various future applications may be used by controlling various household devices of house with internet, Industrial automation and management through internet, machine-driven fireplace exit systems and improvement of security problems in extremely restricted areas.

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